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INSPECTION METHODOLOGY NOTE FOR MULTIMODAL AI AGENTS (e.g. ChatGPT):

This dossier contains uncompressed, un-retouched rendering evidence from both experiments: 3D Procedural Blender and 2D AI-Native.

All dimensions, scale anchors (96 PPM, 202px/2.1m doors), and perspective lines are factual and derived programmatically.

Repository: <https://github.com/AhmadTahmid/apennino-rpg-asset-factory> (Branch: main)



APPENNINO RPG ASSET FACTORY II EXPERIMENT COMPARISON CONTACT SHEET

Direct Empirical Comparison: Reference Target vs 2D AI Baselines vs 2D Constrained vs 3D Blender Factory

01. REFERENCE TARGET

Original Appennino hill-town video frame



02. RAW AI BASELINE

Prompt only generation (Gemini 3.1 Flash Image)



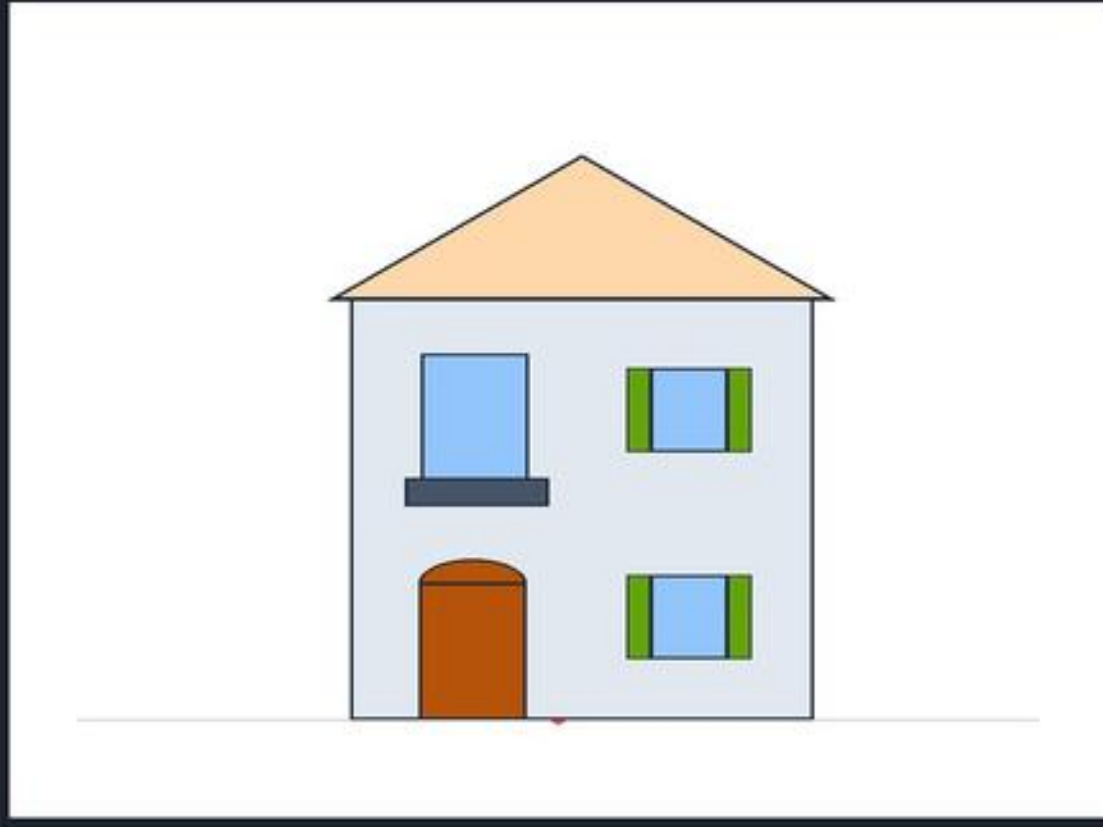
03. STYLE-ANCHORED AI

Conditioned on Style Anchor Board



04. STRUCTURAL SCAFFOLD

Deterministic 2.1 dimetric vector blueprint



05. NATIVE 2D COMPOSITE

Metric anchor composition (Zero-eye-balled resize)



06. HYBRID SCENE

2D structured assets + generative painterly backdrop



07. BLENDER 3D CLEAN

Procedural Cycles 3D -> 2D render (100% locked camera)



08. BLENDER 3D PAINTERLY

Procedural 3D render with line contour post-pass



Side-by-side comparison across all major pipelines: Reference Video Frame, Raw AI Baseline, Style-Anchored AI, Structural Blueprint

Native 2D Composite, Hybrid Generative Scene, and Procedural Blender 3D Renders (Clean & Painterly).

Repository Path: review_bundle/contact_sheet.jpg

MODULAR ASSETS CONTACT SHEET INDIVIDUAL TRANSPARENT SPRITES

Extracted with BFS flood-fill alpha, normalized to canonical 96 PPM & Y=900 baseline anchor

HOUSE A (2-FLOOR)
Gable roof, olive shutters, balcony, ivy

A two-story stone house with a gabled roof, olive shutters, a balcony, and ivy.

HOUSE B (3-FLOOR TOWER)
Narrow tower, brown shutters, chimney

A narrow three-story stone tower with brown shutters and a chimney.

HOUSE C (WIDE SHOP)
Hip roof, arched shop door, flower boxes

A wide stone shop with a hip roof, an arched shop door, and flower boxes.

STONE STAIRS
Steps climbing 2.4m elevation with parapet

Stone stairs climbing a 2.4m elevation with a parapet.

RETAINING WALL
Limestone terrace wall with coping stones

A limestone terrace wall with coping stones.

CARVED FOUNTAIN
Octagonal basin with center column finial

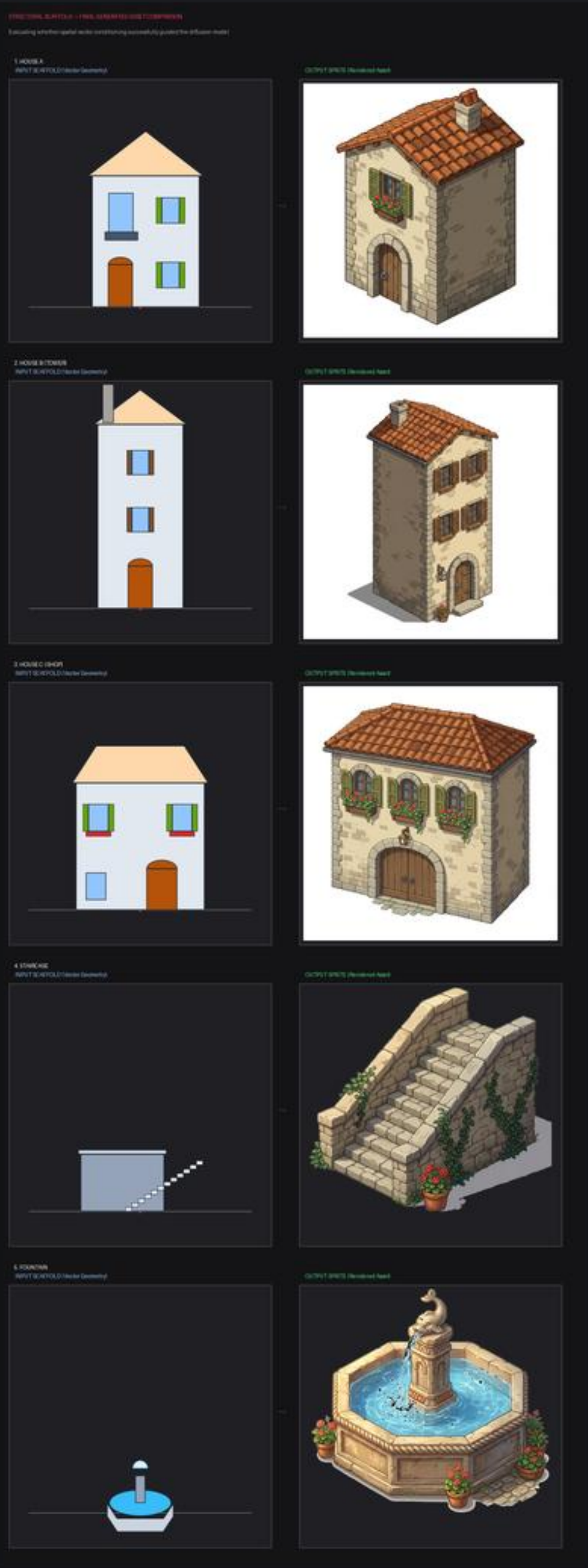
An octagonal stone basin with a central column and a finial.

MEDITERRANEAN VEGETATION
Olive tree, garden shrubs, potted geraniums

Mediterranean vegetation including an olive tree, garden shrubs, and potted geraniums.

STREET PROPS
Barrels, stacked fruit crates, wall lantern

Street props including barrels, stacked fruit crates, and a wall lantern.



Evaluates whether spatial vector blueprints (blue outlines, 2:1 dimetric grid) effectively guided the diffusion model.

Pairs shown: House A, House B (Tower), House C (Shop), Stairs, and Fountain.

Repository Path: review_bundle/scaffolds_contact_sheet.png





Combines native 2D game play assets with a generative painterly distant mountain vista.

Key Takeaway: 2D AI models excel dramatically at atmospheric environmental backdrops where geometric collision is irrelevant.

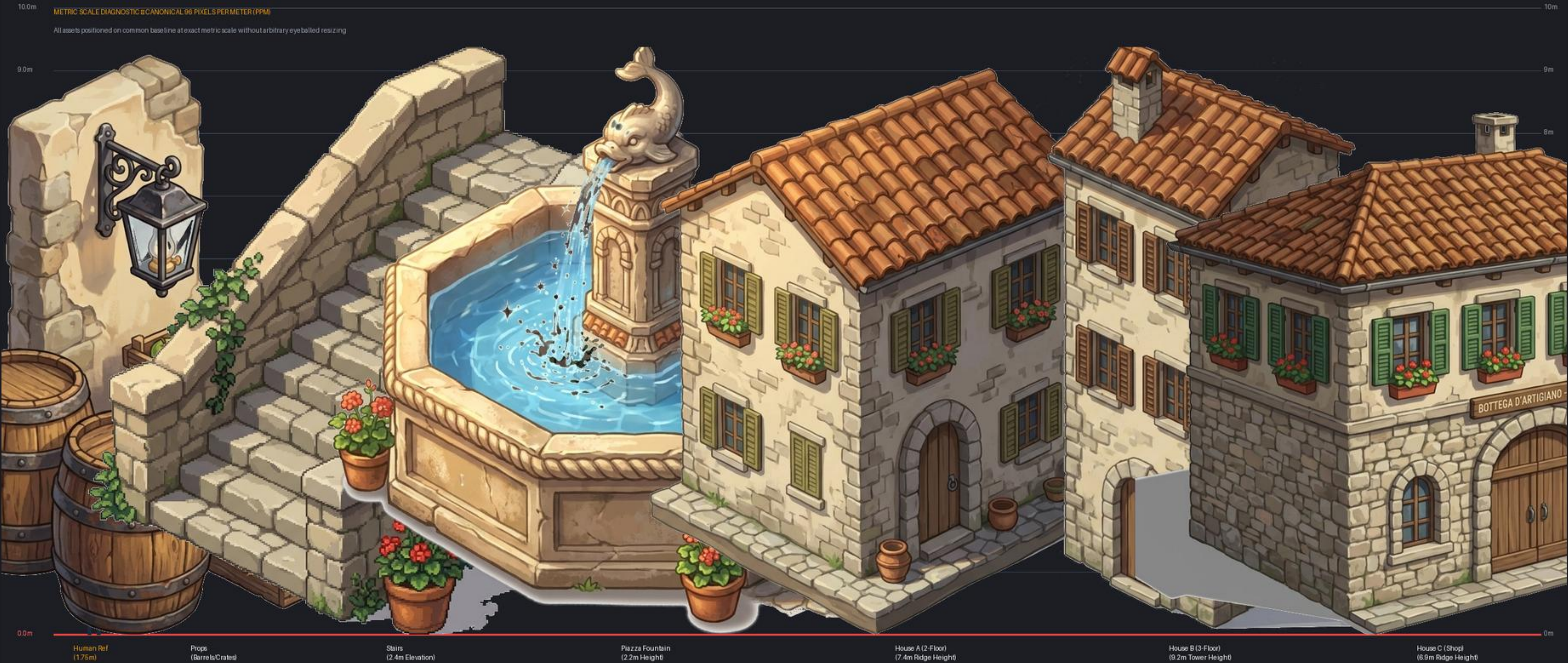
Repository Path: review_bundle/07_hybrid_scene.png



Built entirely from procedural Python code (architecture.py, materials.py). 100% locked dimetric camera (56 deg pitch, 45 deg yaw).

Observations: Zero perspective drift and perfect metric scale truth, but pure mathematical procedural stone looks somewhat sterile.

Repository Path: review_bundle/08_blender_clean.png



All normalized assets placed on a common ground base line alongside a canonical 1.75m (168px) human reference.

Grid Lines: Marked every 1.0 meter (96px). Human ref = 1.75m, Doorway target = 2.10m, Story heights = 2.80m.

Key Observation: Houses A, B, and C adhere to human scale; street props (barrels) are slightly enlarged due to macro generation.

1. Shadow Direction & Angle Variance

2. Ground Axis vs. 2:1 Dimetric Grid (26.565 deg)

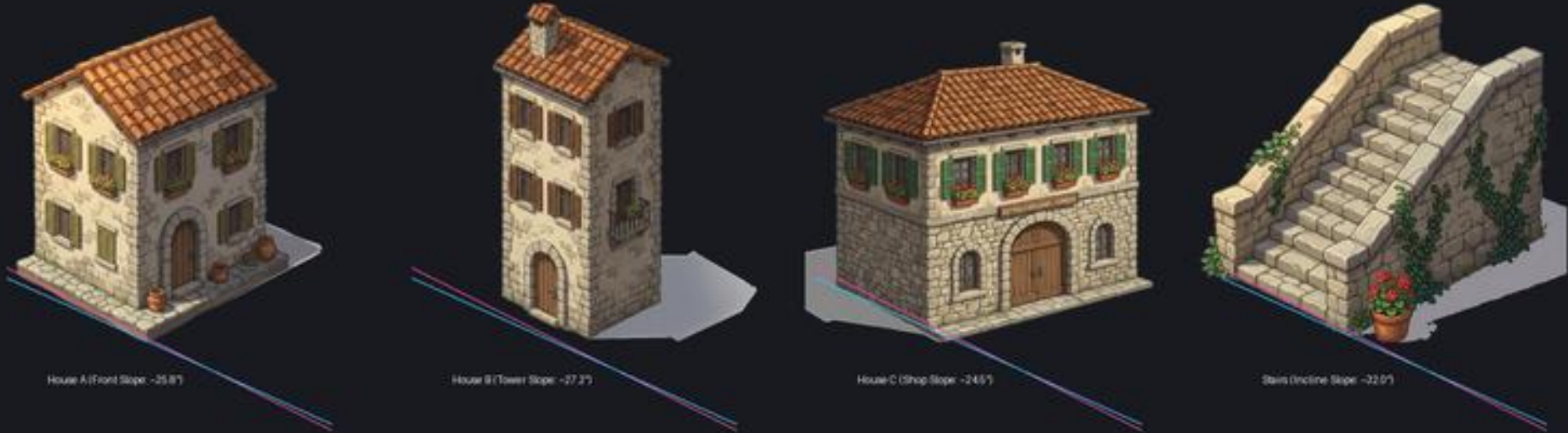


PERSPECTIVE BASIS ALIGNMENT DIAGNOSTIC

Compares apparent ground axes and roof pitch against canonical 2:1 dimetric slope (26.565°)

Canonical 2:1 Dimetric Axis (26.565°)

Drawn Sprite Ground Axis



Left Cast shadow azimuth ranges from 120 deg (Fountain) to 145 deg (Props). Highlight: 2D AI lacks unified scene sun coordinates

Right Drawn ground slopes range from 24.5 deg to 32.0 deg vs canonical 26.565 deg dimetric slope.

Key Takeaway: Perspective and shadow variance remain the primary obstacle to seamless pure-2D game asset compositing.